CryoStack Scientific Cyberinfrastructure

USER EXPERIENCE

Web browser

Documentation · scientific applications · personal workspaces · experiment history

GATEWAY + OPERATIONS

Nginx + aiohttp gateway

HTTP/WebSocket proxying · Jupyter Book · Voilà · static/React routes · application registry

scrypt accounts · GitHub/ORCID · sessions · configurations · workspaces · experiments + events

Control Center · RBAC · audit events · diagnostics · dependency-aware builds · health checks

APPS

CryoLauncher

Modular run planning

models · logs · results

ISSM · Icepack

ICESEE

Ensemble state and

parameter estimation

Filter variants · MPI ensembles

LIVIST

Living Ice Sheet

Temperature explorer

Radar · boreholes · geospatial data

Frozen Legacies

Historical radar catalog

and interpretation tools

Manifests · adapters · GeoJSON

SHARED SERVICES

Application, data, execution, and container abstractions

experiment lifecycle · frontend panels · dataset manifests/adapters · backend-independent results/status

FastAPI relay · connector/direct SSH drivers · Docker publication · Spack/Apptainer environments

EXECUTION

Local

Python · notebooks · MPI

Interactive and lightweight workflows

Remote HPC

Direct SSH or outbound connector

rsync · archives · Slurm · logs · outputs

AWS Batch

S3 · IAM · ECR · Fargate (default)

Fargate · EC2 (selected workflows)

Reproducible environments and artifacts: Spack locks · Docker/Apptainer images · run records · data catalogs